User manual for conducting experiments with the developed measurement device

1 Preparation Before the Experiment

1. Prepare the plant-based meat sample that will be put inside of the measurement device.
2. Set the room temperature to be constant using air conditioning (for example to 24 ℃ in winter).
3. Check the output voltage of the DC power supply (HX711 based weight scale module: 5.0 V, light source(0.5 A / 4.8 V halogen lamp): 4.8 V).
4. Check the wiring connections (refer to the circuit diagram on the last page).

2 Scale calibration (Optional)

1. Turn on the power of the Raspberry Pi, the scale circuit IC and the light source.
2. Use Geany to launch `hx711_scale_calibration.py` (refer to Programs/Other Programs of the same depository).
3. Place an object with a weight that is known(for example a coin) on the scale and enter it's weight in the terminal (this number has to be an integer). Record the value of the "Reference Unit" shown in the terminal.
4. Use Geany to open `hx711_scale.py` (refer to Programs/Other Programs of the same depository) and change the value of the "Reference Unit" to the one acquired in the previous step.
5. Run `hx711_scale.py`, place the plant-based meat sample on the scale, and record its initial mass before conducting the experiment.

3 Pre-experiment Setup of Developed Capture Software(`capture.py`)

3.1 Camera Settings

1. Use Geany to launch `capture.py` (listed in Programs/Capture Software).
2. Enter the camera settings in the first tab.
   - Shutter speed: (26.5 μs $\sim$ 112 s)
   - Lens Position (0.0 $\sim$ 10.0)
   - Colour gains [Red, Blue]: (0.0 $\sim$ 32.0, 0.0 $\sim$ 32.0)
   - Analog Gain: (1.0 $\sim$ 16.0)
3. Enter the light source settings
   - Lamp Drive: Select the pin for the light source (near-infrared light source) (GPIO4 $\sim$ GPIO27)
   - Pre-heat: Enter the pre-heat time (s)

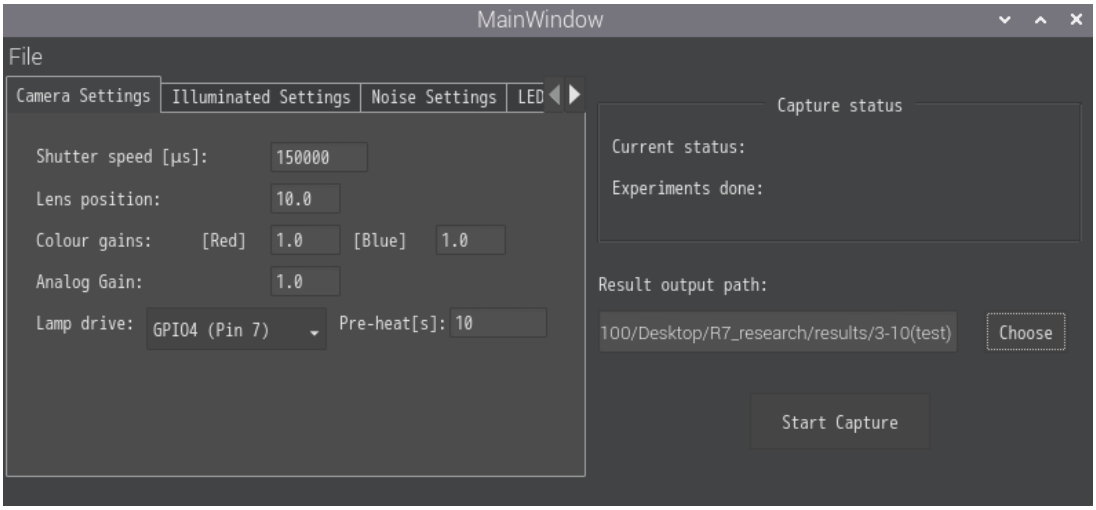


Figure 1: Camera Settings Tab

3.2 Illuminated Settings

1. Enter the capture settings for when the light source (near-infrared light source) is turned on.

   - Capture NIR illuminated images: Select whether to capture images with the NIR light source turned on (optional).
   - Illuminated image count: Number of images to capture while the light source is on.
   - Pause between capture: Waiting time between captures (s).
   - Save raw images: Select whether to save raw images (DNG format) (optional).
   - Illuminated image name: Image name (saved in the format "image name + image number").

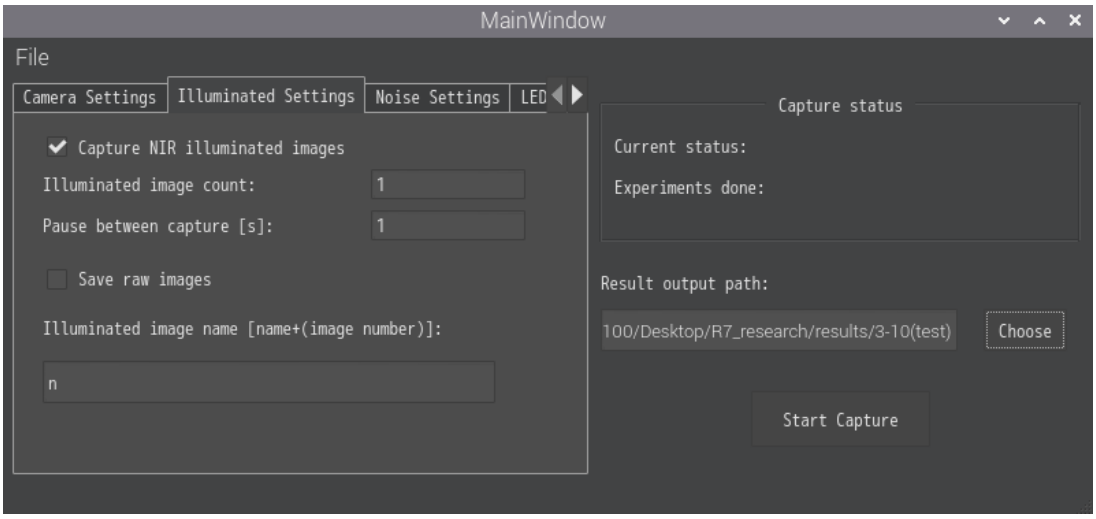


Figure 2: Illuminated Settings Tab

3.3 Noise Settings

1. Enter the capture settings when the light source is turned off (noise capture).

   - Capture noise images: Select whether to capture images with the NIR light source turned on (optional).
   - Noise image count: Number of images to capture.
   - Pause between capture: Waiting time between captures (s).
   - Save raw image: Select whether to save raw images (DNG format) (optional).
   - Noise image name: Image name (saved in the format "image name + image number").

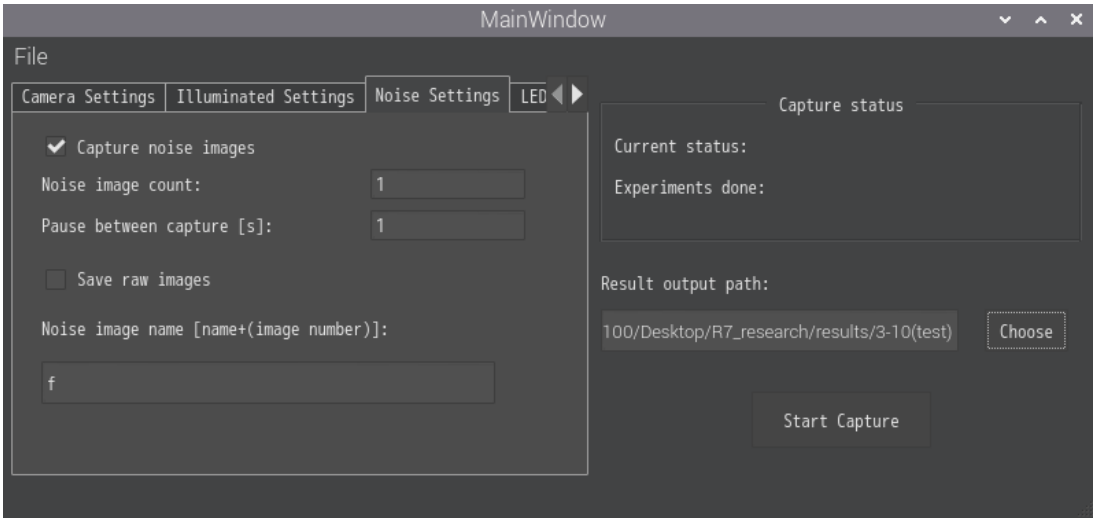


Figure 3: Noise Settings Tab

3.4 LED Settings (Optional)

1. Enter the capture settings when the LEDs are turned on.

   - LED illuminated capture checkbox: Select whether to capture images when the LEDs are turned on (optional).
   - LED1, LED2: Select the pin number for each LED (GPIO4 $\sim$ GPIO27).
   - Illuminated image count: Number of images to capture.
   - Pause between capture: Waiting time between captures (s).
   - Save raw image: Select whether to save raw images (DNG format) (optional).
   - LED illuminated image name: Image name (saved in the format "image name + image number").

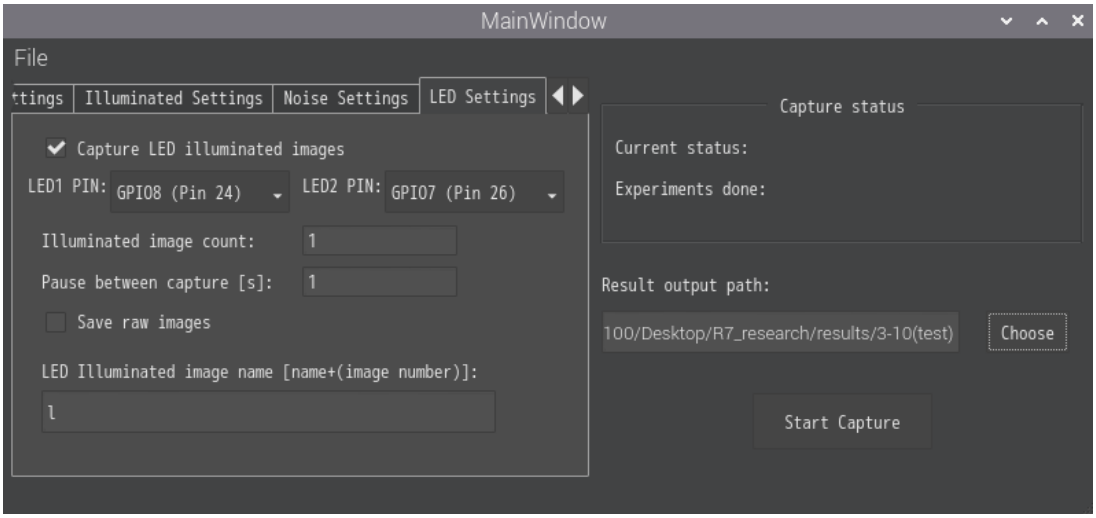


Figure 4: LED Settings Tab

3.5 Experiment Settings

1. Enter the advanced settings for experiment capture。

   - Get metadata of each capture: Select whether to save metadata for each capture (camera settings, actual values, camera temperature, etc.) (optional).
   - Single Shutter Speed / Multiple Shutter Speed: Select whether to perform the experiment with a single or multiple exposure times. If multiple is selected, enter the range in μs (from the exposure time in From to the exposure time in To, varying with the step size specified by By).
   - Number of experiments: Specify the number of experiments. One experiment consists of a sequence of operations: mass measurement, noise imaging, LED illuminated imaging, light source pre-heating, and imaging with the light source turned on.
   - Time between experiments (TBE): Specify the waiting time between experiments (min). The actual waiting time is "TBE −pre-heat time".
   - Result output path: The destination for saving each image, the metadata of each image, and the mass measurement results.

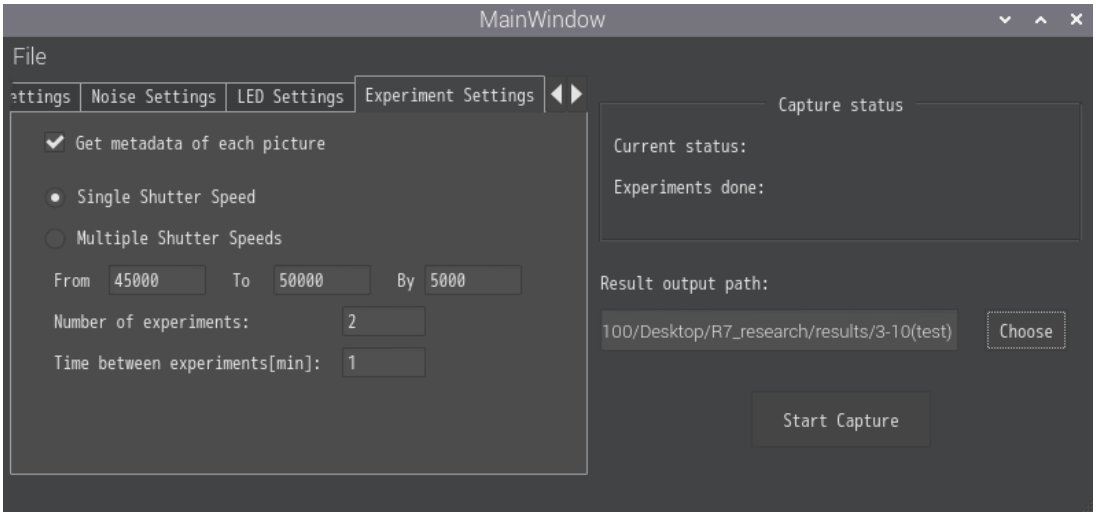


Figure 5: Experiment Settings Tab

3.6 Scale Settings (Optional)

1. Enter the scale settings.

   - Weigh the sample during experiments: Select whether to measure the mass of the sample during the experiments (optional).
   - DAT PIN, CLK PIN: Select each pin of the measurement module (GPIO4 $\sim$ GPIO27).
   - Reference Unit: Enter the value obtained in Section 2.
   - Number of measurements: Enter how many measurement results will be used to calculate the average value.

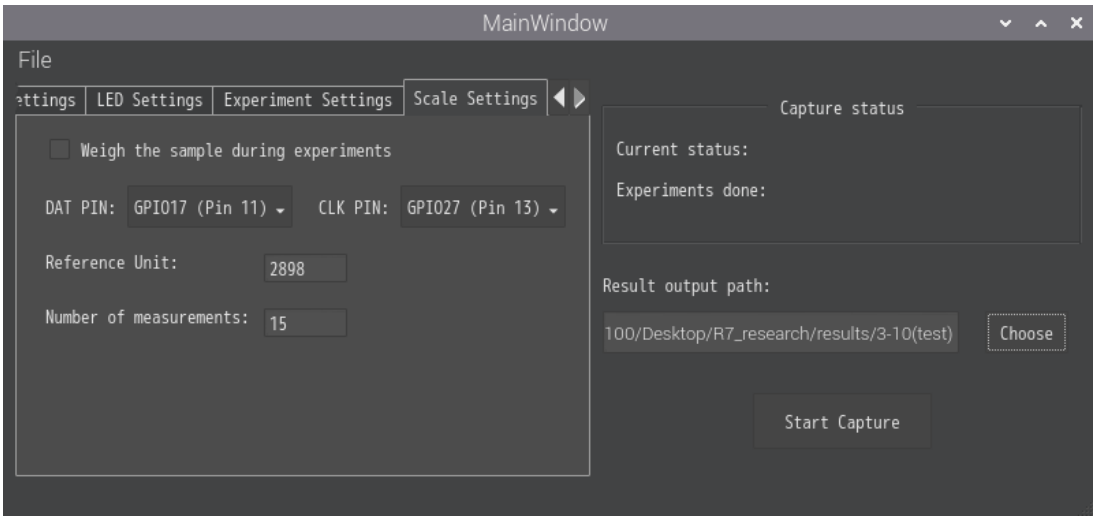


Figure 6: Scale Settings Tab

4 Experimental Procedure

1. Darken the test room (close the blinds, reduce the monitor brightness, etc.).
2. Place the sample on the scale and attach the plastic film cover.
3. Set the light source at an appropriate angle and fix it in place.
4. Cover the entire device with cardboard and orient the monitor in the opposite direction from the device.
5. Configure the various settings in `capture.py`.
6. Press the "Start Capture" button.

5 After the Experiment

1. Exit `capture.py`.
2. Remove the plant-based meat sample from the device and measure its mass again using `hx711_scale.py` (obtain the final mass m_f).
3. Turn off the DC power supplies used.
4. Save the data to a USB flash drive or any other memory device.
5. Turn off the Raspberry Pi(if data analysis is done on a regular computer).

6 Image Processing Procedure Using `image.py`

1. Use Geany to launch `image.py` (refer to Programs/Image Analysis Software)
2. Click "Open File" and select one image from the initial time when the near-infrared light source is turned on.

6.1 Blur and Thresholding

1. Select "Gaussian blur" from "Blur Method" and select "Thresh Binary" from "Thresholding Method".
2. Set "Kernel" to (3, 3) and set "signalX" to 0.
3. Adjust the "Threshold Value" until the shape of the plant-based meat sample is clearly displayed, and take note of the value. Each time the settings are changed, click "Apply Settings" to update the image displayed on the right.

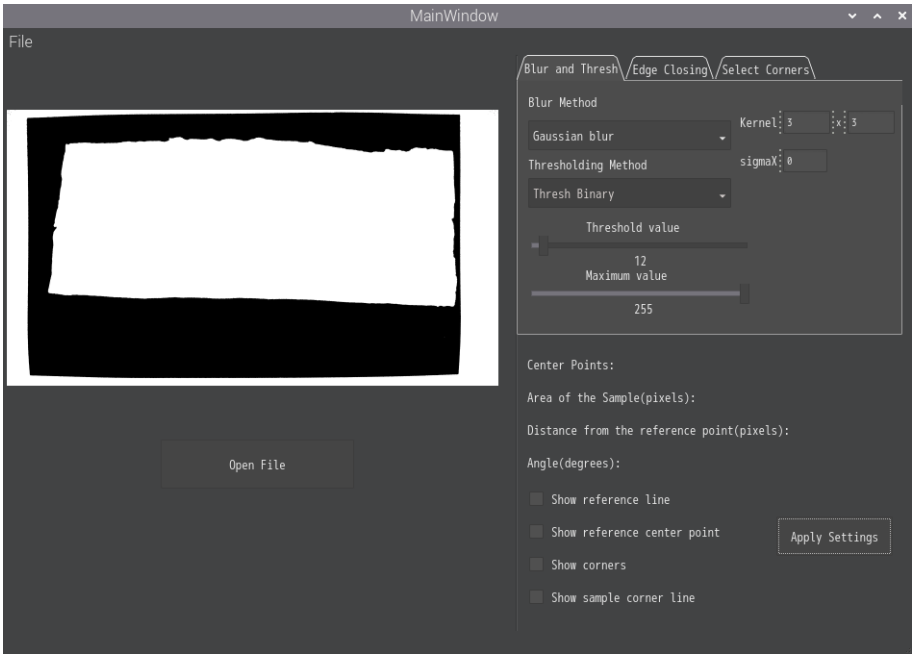


Figure 7: Blur and Thresh Tab

6.2 Edge Closing

1. Select “Edge Closing” and set “Kernel” to (3, 3).
2. Click the checkbox for “Area and Center Point Detection”.
3. Adjust “min_area” and “max_area” until the contour of the sample region is displayed with a yellow line, and take note of the values.
4. Enter the recorded values into another program and perform multiple image processing (calculation of APV and API for all images) (for details, see https://github.com/kostasb100/low-cost-plant-based-meat-moisture-device).

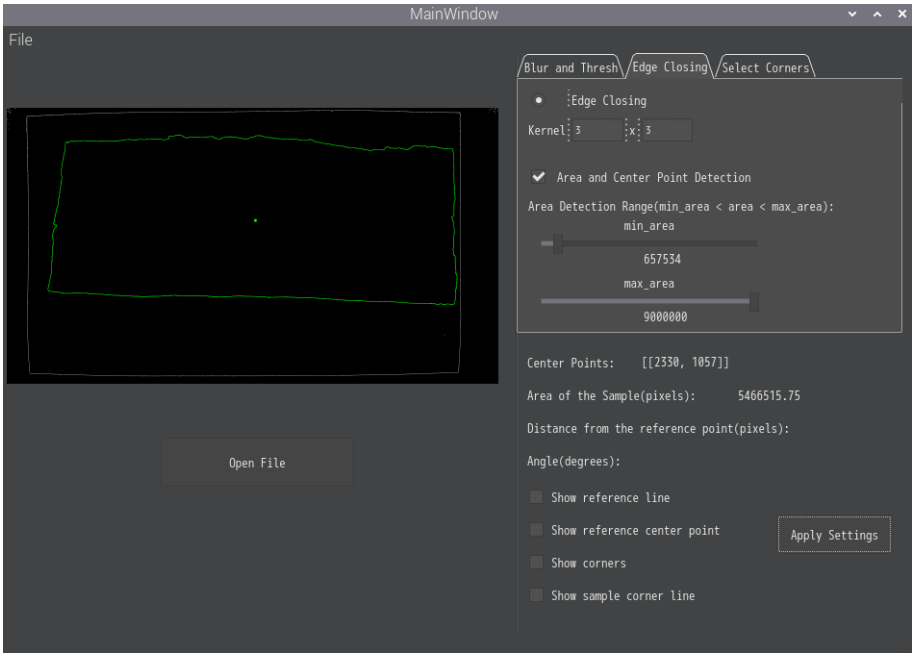


Figure 8: Edge Closing Tab

7 Troubleshooting

During the use of this device, there may be cases where the light sources (LEDs, near-infrared light source) do not turn on. In such cases, check the following steps.

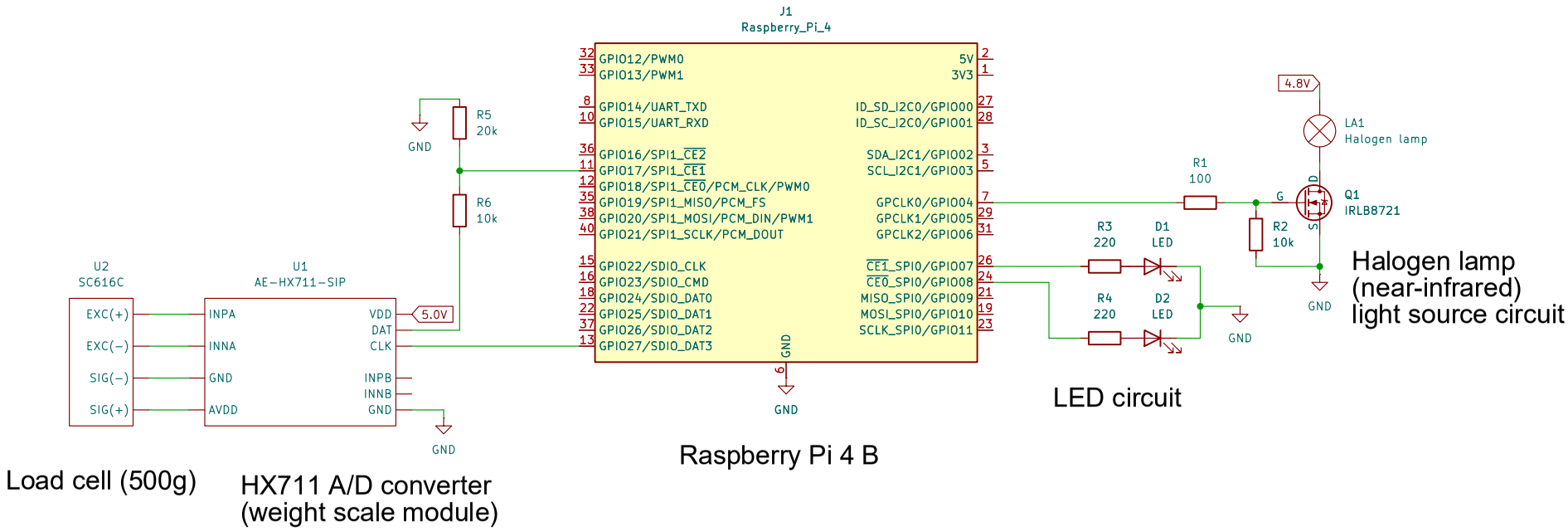
7.1 Hardware

1. Using a multimeter check whether the power for every circuit(scale, infrared light source) is being delivered at the correct value. For LED circuits, check whether the pins are connected correctly.
2. Turn off the device and check the wiring (especially the wiring inside the device).
3. Try activating the near-infrared light source (LED circuits) using an external signal (a voltage source other than the Raspberry Pi GPIO pins).
4. For further details, refer to https://github.com/kostasb100/low-cost-plant-based-meat-moisture-device.

7.2 Software

This software may stop just before capturing raw images. In such cases, check the following steps.

1. Execute `sudo apt update && sudo apt upgrade` in the terminal.
2. Restart the Raspberry Pi.
3. For further details, refer to https://github.com/kostasb100/low-cost-plant-based-meat-moisture-device.



Sheet: /
File: tik_halogen_kairo.kicad_sch

Title: Moisture Measurement Device Circuit

Size: A4	Date: 2026−02−16	Rev:
KiCad E.D.A. 9.0.0		Id: 1/1